

Due: Thursday 10/30 at 11:59pm.

Policy: Can be solved in groups (acknowledge collaborators) but must be submitted individually.

Make sure to show all your work and justify your answers.

Note: This is a typical exam-level question. On the exam, you would be under time pressure, and have to complete this question on your own. We strongly encourage you to first try this on your own to help you understand where you currently stand. Then feel free to have some discussion about the question with other students and/or staff, before independently writing up your solution.

Note: Leave the self-assessment sections blank for the original submission of your homework. After the homework deadline passes, we will release the solutions. At that time, you will review the solutions, self-assess your initial response, and complete the self-assessment sections below. The deadline for the self-assessment is 1 week after the original submission deadline.

Your submission on Gradescope should be a PDF that matches this template. Each page of the PDF should align with the corresponding page of the template (page 1 has name/collaborators, question begins on page 2.) **Do not reorder, split, combine, or add extra pages.** The intention is that you print out the template, write on the page in pen/pencil, and then scan or take pictures of the pages to make your submission. You may also fill out this template digitally (e.g. using a tablet.)

First name	
Last name	
SID	
Collaborators	

Q1 Bayes Nets**(24 points)**

Consider the following Bayes Net that has four variables E , S , M , and B . The associated probabilities are also given in the following tables.

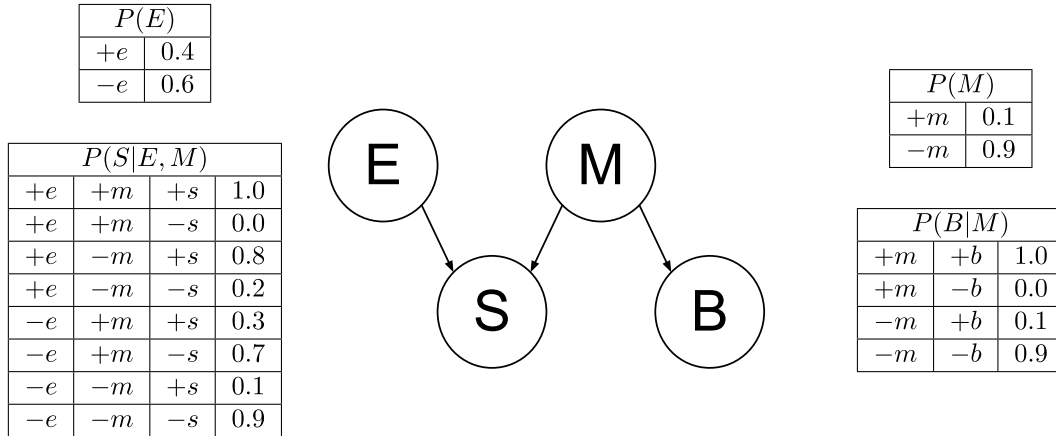


Figure 1: Bayes Net and probability tables.

In the following five questions, do not just answer with a numerical value. Write explicitly what probability expression you are calculating. For instance you should write $P(A)P(B) = 0.2$ instead of just 0.2.

Q1.1 (2 points) Write an expression to compute the value of the joint distribution $P(-e, -s, -m, -b)$. What is its value?

Q1.2 (2 points) Write an expression to compute the value of $P(+b)$. What is its value?

Q1.3 (2 points) Write an expression to compute the value of $P(+m \mid +b)$. What is its value?

(Question 1 continued...)

Q1.4 (2 points) Write an expression to compute the value of $P(+m \mid +s, +b, +e)$. What is its value?

Q1.5 (2 points) Write an expression to compute the value of $P(+e \mid +m)$. What is its value?

Q1(a) Self-Assessment – leave this section blank for your original submission. We will release the solutions for this problem after the deadline for this assignment has passed. After reviewing the solutions for this problem, assess your initial response by checking one of the following options:

- ☐ I fully solved the problem correctly, including fully correct logic and sufficient work (if applicable).
- ☐ I got part or all of the question incorrect.

If you selected the second option, explain the mistake(s) you made and why your initial reasoning was incorrect (do not re-iterate the solution. Instead, reflect on the errors in your original submission). Approximately 2-3 sentences for each incorrect sub-question.

(Question 1 continued...)

Now consider the following Bayes Net in which all variables are **binary**. We want to compute the query $P(B, D \mid +f)$, so we run variable elimination with the variable elimination ordering being A, C, E, G .

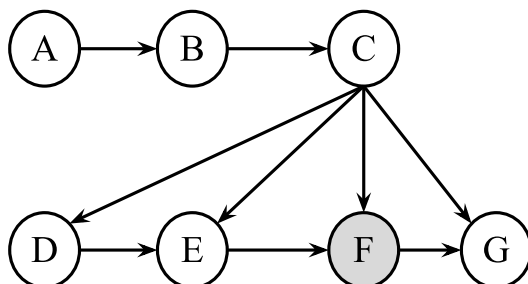


Figure 2: Bayes Net.

After observing evidence on F , we have the following factors:

$$P(A), P(B \mid A), P(C \mid B), P(D \mid C), P(E \mid C, D), P(+f \mid C, E), P(G \mid C, +f)$$

When we eliminate the variable A , we then create a new factor $f_1(B) = \sum_a P(B \mid a) \cdot P(a)$ and the remaining factors are:

$$f_1(B), P(C \mid B), P(D \mid C), P(E \mid C, D), P(+f \mid C, E), P(G \mid C, +f).$$

Q1.6 (2 points) When we eliminate C next, what is the new factor f_2 we obtain? Furthermore, list the leftover factors.

Q1.7 (2 points) When we eliminate E next, what is the new factor f_3 we obtain? Furthermore, list the leftover factors.

Q1.8 (2 points) When we eliminate G next, what is the new factor f_4 we obtain? Furthermore, list the leftover factors.

(Question 1 continued...)

Q1.9 (2 points) How can we compute $P(B, D \mid + f)$ using the factors from Q1.8? Explain in a couple of sentences.

Q1.10 (2 points) Between f_1 , f_2 , f_3 , and f_4 , which is the largest factor (or equivalently whose table has the most rows)?

Q1.11 (4 points) For the same query $P(B, D \mid + f)$, find a new variable elimination ordering that minimizes the size of the largest factor created. Fill in your ordering and generated factors below. Hint: The largest factor should have only 2 variables.

Variable Eliminated	Factor Generated

(Question 1 continued...)

Q1(b–c) Self-Assessment – leave this section blank for your original submission. We will release the solutions for this problem after the deadline for this assignment has passed. After reviewing the solutions for this problem, assess your initial response by checking one of the following options:

- ☐ I fully solved the problem correctly, including fully correct logic and sufficient work (if applicable).
- ☐ I got part or all of the question incorrect.

If you selected the second option, explain the mistake(s) you made and why your initial reasoning was incorrect (do not re-iterate the solution. Instead, reflect on the errors in your original submission). Approximately 2-3 sentences for each incorrect sub-question.